

What if Mass Eats the Vacuum?

David Plotz

May 13, 2026

Abstract

This paper introduces a classical field theory of gravity built on the Maxwell equations and a hypothesis: that mass absorbs zero-point radiation, inducing a net inward momentum flow. We construct a toy model of a massive particle, demonstrate energy and momentum conservation, and show that the three classic tests of general relativity (Mercury's precession, gravitational lensing, and gravitational redshift) are reproduced to leading order. We identify where the theory diverges from general relativity at higher order. This approach provides a Lorentz-invariant classical alternative to general relativity, with implications explored in the final sections.

1 Introduction

Modern physics rests on two towering achievements: the Standard Model, underpinned by the Dirac equation, and General Relativity, based on Einstein's field equations. One governs the small, the other the large. Each is remarkably successful in its domain.

These two frameworks, however, are incompatible. When applied simultaneously, the equations break down and yield infinities. Proposals like string theory offer elegant mathematics but few concrete predictions. Despite nearly a century of effort, no unifying framework has emerged. A fresh approach is warranted.

The scientific method permits us to begin with a hypothesis. Newton postulated an inverse square force law without explaining why gravity falls off with distance. Maxwell wrote down his equations before anyone understood the photon. A hypothesis does not need to come with a microscopic derivation. A theory only needs to make testable predictions. That is the standard we hold ourselves to here.

This paper proposes such an approach. Rather than quantizing Einstein's field equations or geometrizing the Dirac equation, we suggest something simpler and stranger: what if mass eats the vacuum? The consequences, as we will see, extend beyond gravity.

1.1 Roadmap

This paper is structured as a straightforward application of the scientific method: we begin with a hypothesis, outline the assumptions that support it, explore its mathematical consequences, compare its predictions to experiment, and finally examine whether it motivates a deeper conceptual shift.

In **Section 2**, we introduce the hypothesis: that mass acts as a sink for radiation and, in particular, as a sink for a pervasive stochastic zero-point radiation field (Z.P.R.). We first present the idea intuitively, anchoring the theory in a physical picture, and then spell it out mathematically.

In **Sections 3 and 4**, we lay out the two assumptions underlying the theory. By “assumptions,” we mean elements already present in the scientific literature, perhaps unconventional but not novel to this work. **Section 3** introduces the stochastic zero-point radiation (Z.P.R.) field as the physical manifestation of the vacuum. **Section 4** adopts the Maxwell equations as the correct mathematical framework for describing gravity.

Sections 5 and 6 contain the technical core of the paper. We introduce a toy model of a massive particle and discuss characteristics of the toy model. We examine whether the inflow of radiation can stabilize mass against gravitational self-energy and show that the energy needed for stability of our toy model is less than that available in the Z.P.R. We also demonstrate conservation of energy and momentum.

In **Section 7**, we test the theory against observation. We demonstrate that it reproduces the three classic predictions of general relativity to leading order:

perihelion precession, gravitational lensing, and redshift. We also identify where the theory diverges from general relativity, suggesting a possible observational test.

Section 8 explores a surprising result: when the field equations are rewritten in spinor form, they yield a structure identical to the Dirac equation, without quantization. And finally, **Section 9** concludes with a discussion of open questions, potential avenues for further investigation, and the broader implications of the framework proposed here.

2 The Hypothesis

In this section, we introduce the hypothesis of this paper. Before proceeding, we clarify the distinction between assumptions and hypothesis as used here:

The scientific method begins with a **hypothesis**. We make one here. This is distinct from our two **assumptions**, which are drawn from existing theoretical frameworks. The hypothesis is the novel, testable claim of this paper.

Here we first present the hypothesis intuitively, anchoring the theory in a physical picture, and then spell it out mathematically.

2.1 A Physical Picture: Mass as a Radiation Sink

Before introducing equations, we present a simple physical picture to guide intuition. Consider a neutral, perfectly conducting iron sphere, heated until it glows red. When placed in the vacuum of space, it emits infrared radiation and cools down, approaching absolute zero.

But at zero Kelvin, both classical and quantum theories suggest the presence of a background electromagnetic field called the stochastic zero-point radiation (Z.P.R.) field. A fuller treatment of the Z.P.R. will be offered in **Section 3**, but for now, it is enough to imagine a low-level radiation noise present even in the coldest vacuum. The iron sphere, after it has cooled down and come into equilibrium with the vacuum, then absorbs radiation from this ambient field.

The hypothesis proposed here is that mass absorbs radiation from the zero-point field even at absolute zero, producing a net inward flow of momentum toward the mass.”

The remainder of the paper builds from this intuition and develops the consequences of this idea.

2.2 Hypothesis: The Math

The hypothesis of this paper is that mass creates an inward-pointing spectral momentum density:

$$\boxed{\mathbf{P}(r) = -\hat{\mathbf{r}} \frac{Gm\hbar}{r^2}} \tag{1}$$

where $\mathbf{P}(r)$ is the momentum density per mode in k -space.

A photon traveling through this field experiences a cumulative momentum shift:

$$\Delta \mathbf{p} = \int_{\text{path}} \mathbf{P}(r) d\phi \quad (2)$$

where $d\phi = k dr$ is the infinitesimal phase along the path, so that the momentum transfer accumulates per cycle of oscillation.

The intuition underlying this hypothesis has respectable precedent. Feynman notes that light “falls” in a gravitational field [1], which is precisely the picture developed here: mass acts as a sink for radiation. We do not specify the microscopic mechanism by which mass couples to the ZPR field. The implications of this hypothesis are tested in **Section 7**.

3 Zero Point Radiation (ZPR)

We now introduce our first assumption: the **stochastic zero-point radiation field**. “Stochastic” means random, in a way that will be clearly defined below, “zero-point” refers to zero-temperature, and “radiation” in this context simply means light.

Historically, the Michelson–Morley experiment aimed to detect the luminiferous aether, the medium through which light was thought to propagate. The null result helped solidify the notion of a truly empty vacuum. In contrast, the stochastic zero-point radiation field treats the vacuum as filled with fluctuating electromagnetic fields, even at absolute zero.

This theory was first developed by T.H. Boyer in the 1960s and has since been extended by Boyer and others under the term stochastic electrodynamics (SED); see [2] for a comprehensive review. Among its successes, SED has been shown to reproduce the Planck blackbody spectrum without quantum assumptions [3], the harmonic oscillator ground-state energy $\frac{1}{2}\hbar\omega$, and the Casimir and van der Waals forces, among other results.

Taken together, these results support the view that the vacuum is an active medium, filled with fluctuating modes that persist even at $T = 0$. This is not the mainstream view. We adopt it on the grounds that it has a substantial body of supporting results and, as we will show, leads to a consistent and predictive theory of gravity. We take no position on whether SED can serve as a complete replacement for quantum mechanics; we use it here as a classical framework well-suited to the problem at hand.

We therefore assume that the vacuum is filled with a classical, Lorentz-invariant random electromagnetic radiation field even at absolute zero, carrying a spectral energy density:

$$\rho(\omega) = \frac{1}{2} \hbar \omega \quad (3)$$

3.1 Mathematical Definitions of the Z.P.R.

The Z.P.R. field is stochastic and Gaussian. The electric and magnetic fields, in natural units ($c = 1$), are defined as:

$$\begin{aligned}\mathbf{E}(\mathbf{x}, t) &= \sum_{\lambda=1}^2 \int d^3k \hat{\epsilon}(\mathbf{k}, \lambda) \sqrt{\frac{\hbar\omega}{8\pi^2}} W_{\lambda}(\mathbf{k}) e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} + \text{c.c.} \\ \mathbf{B}(\mathbf{x}, t) &= \sum_{\lambda=1}^2 \int d^3k \frac{\mathbf{k} \times \hat{\epsilon}(\mathbf{k}, \lambda)}{|\mathbf{k}|} \sqrt{\frac{\hbar\omega}{8\pi^2}} W_{\lambda}(\mathbf{k}) e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} + \text{c.c.}\end{aligned}\tag{4}$$

where $\hat{\epsilon}(\mathbf{k}, \lambda)$ are polarization vectors satisfying:

$$\hat{\epsilon} \cdot \mathbf{k} = 0, \quad \hat{\epsilon}_1 \cdot \hat{\epsilon}_2 = 0, \quad |\hat{\epsilon}_{\lambda}| = 1\tag{5}$$

and $W_{\lambda}(\mathbf{k})$ are complex Gaussian random variables with correlation properties:

$$\begin{aligned}\langle W_{\lambda}(\mathbf{k}) \rangle &= 0 \\ \langle W_{\lambda}(\mathbf{k}) W_{\lambda'}^*(\mathbf{k}') \rangle &= \delta_{\lambda\lambda'} \delta^3(\mathbf{k} - \mathbf{k}') \\ \langle W_{\lambda}(\mathbf{k}) W_{\lambda'}(\mathbf{k}') \rangle &= 0\end{aligned}\tag{6}$$

where $\langle \cdot \rangle$ denotes the stochastic average over the ensemble of random field realizations. This background field is homogeneous, isotropic, and statistically stationary, with the full ensemble remaining invariant under Lorentz transformations.

3.2 Derivation of Energy Spectrum

We now verify that this field definition gives the correct energy spectrum. We need to evaluate $\langle \mathbf{E}^2 \rangle$ and $\langle \mathbf{B}^2 \rangle$. Let us focus on $\langle \mathbf{E}^2 \rangle$ as the calculation for the magnetic field is identical.

Write $\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_+^*$ where $\mathbf{E}_+$ is

$$\mathbf{E}_+(\mathbf{x}, t) = \sum_{\lambda=1}^2 \int d^3k \hat{\epsilon}(\mathbf{k}, \lambda) \sqrt{\frac{\hbar\omega}{8\pi^2}} W_{\lambda}(\mathbf{k}) e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}\tag{7}$$

Then:

$$\mathbf{E}^2 = (\mathbf{E}_+ + \mathbf{E}_+^*)^2 = \mathbf{E}_+^2 + 2\mathbf{E}_+ \cdot \mathbf{E}_+^* + (\mathbf{E}_+^*)^2\tag{8}$$

Taking the stochastic average:

$$\langle \mathbf{E}^2 \rangle = 2\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle\tag{9}$$

The $\mathbf{E}_+^2$ and $(\mathbf{E}_+^*)^2$ terms vanish because $\langle WW \rangle = 0$, by equation (6). Evaluating the remaining term:

$$\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle = \sum_{\lambda, \lambda'} \int d^3k d^3k' \hat{\epsilon}_\lambda \cdot \hat{\epsilon}_{\lambda'}^* \frac{\hbar \sqrt{\omega \omega'}}{8\pi^2} \langle W_\lambda(\mathbf{k}) W_{\lambda'}^*(\mathbf{k}') \rangle e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} e^{-i(\mathbf{k}' \cdot \mathbf{x} - \omega' t)} \quad (10)$$

Using the correlation function $\langle W_\lambda(\mathbf{k}) W_{\lambda'}^*(\mathbf{k}') \rangle = \delta_{\lambda\lambda'} \delta^3(\mathbf{k} - \mathbf{k}')$:

$$\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle = \sum_{\lambda} \int d^3k |\hat{\epsilon}_\lambda|^2 \frac{\hbar \omega}{8\pi^2} \quad (11)$$

Since $\sum_{\lambda=1}^2 |\hat{\epsilon}_\lambda|^2 = 2$ (two orthogonal unit polarization vectors):

$$\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle = \int d^3k \frac{\hbar \omega}{4\pi^2} \quad (12)$$

Therefore:

$$\langle \mathbf{E}^2 \rangle = 2 \int d^3k \frac{\hbar \omega}{4\pi^2} \quad (13)$$

By identical calculation:

$$\langle \mathbf{B}^2 \rangle = 2 \int d^3k \frac{\hbar \omega}{4\pi^2} \quad (14)$$

The total energy density is:

$$\langle u \rangle = \frac{1}{8\pi} (\langle \mathbf{E}^2 \rangle + \langle \mathbf{B}^2 \rangle) = \frac{1}{8\pi} \cdot 4 \int d^3k \frac{\hbar \omega}{4\pi^2} = \frac{1}{(2\pi)^3} \int d^3k \hbar \omega \quad (15)$$

Since $d^3k/(2\pi)^3$ represents the density of states in momentum space (the number of modes per unit volume in k -space), we can interpret this as:

$$\langle u \rangle = \int \frac{d^3k}{(2\pi)^3} \hbar \omega \quad (16)$$

This confirms that each electromagnetic normal mode (specified by wavevector $\mathbf{k}$ and polarization λ) carries energy $(1/2)\hbar\omega$ in the zero-point radiation field:

$$\rho(\omega) = \frac{1}{2} \hbar \omega \quad (17)$$

4 Maxwell Equations for Gravity and Their Solutions

In this section we introduce the second of our two assumptions. The idea that gravity obeys equations analogous to those of electromagnetism was first proposed by Oliver Heaviside in 1893 [4], predating general relativity. Specifically, we assume that gravity obeys the Maxwell equations with mass currents replacing charge and charge currents as sources. In natural units ($c = 1$):

$$\begin{aligned}\nabla \times \mathbf{B} - \partial_t \mathbf{E} &= -4\pi G \mathbf{J}, & \nabla \times \mathbf{E} + \partial_t \mathbf{B} &= 0 \\ \nabla \cdot \mathbf{E} &= -4\pi G \rho, & \nabla \cdot \mathbf{B} &= 0\end{aligned}\tag{18}$$

These equations describe a vector field sourced by mass density ρ and mass current $\mathbf{J}$, with G the gravitational coupling constant. Note that $\mathbf{E}$ here is a gravitational acceleration field (force per unit mass), analogous to the electric field being force per unit charge.

The continuity equation, although derivable from the above, is worth stating explicitly:

$$\nabla \cdot \mathbf{J} + \partial_t \rho = 0\tag{19}$$

The presence of $\mathbf{B}$, a gravitomagnetic field, may be unexpected. It emerges naturally: just as a moving charge generates a magnetic field, a moving mass generates a gravitomagnetic field. The effect is negligible unless the mass is both extremely large and rapidly moving. **Section 7** discusses a physical manifestation and its potential testability.

We note that Maxwell-like equations for gravity also arise in gravitoelectromagnetism (GEM), the linearized weak-field approximation to general relativity [5]. The present approach differs in a key respect: GEM is derived as an approximation, valid only in the weak-field, slow-motion limit, and its equations are not Lorentz invariant. Here, we take the Maxwell equations as fundamental and exact, inheriting full Lorentz invariance by construction. The two frameworks agree at leading order, but differ at higher order (see Section 7) and in their deeper structure (see Section 8).

The remainder of this section develops mathematical tools that will be used throughout the paper.

4.1 Irrotational and Solenoidal Decomposition

Since any vector field can be split into irrotational (curl-free) and solenoidal (divergence-free) components,

$$\mathbf{V} = \mathbf{V}_I + \mathbf{V}_S, \quad \nabla \cdot \mathbf{V}_S = 0, \quad \nabla \times \mathbf{V}_I = 0\tag{20}$$

We may apply this to our gravitational fields yielding two decoupled subsystems:

Static Fields (Irrotational):

$$\nabla \cdot \mathbf{E}_I = -4\pi G \rho, \quad \partial_t \mathbf{E}_I = 4\pi G \mathbf{J}_I, \quad \nabla \cdot \mathbf{J}_I + \partial_t \rho = 0\tag{21}$$

Radiation Fields (Solenoidal):

$$\nabla \times \mathbf{B}_S - \partial_t \mathbf{E}_S = -4\pi G \mathbf{J}_S, \quad \nabla \times \mathbf{E}_S + \partial_t \mathbf{B}_S = 0\tag{22}$$

This decomposition reflects a meaningful physical distinction between the **static gravitational attraction** and the **propagating radiation** field. In

the rest frame of the mass, the irrotational and solenoidal subsystems decouple. The irrotational equations govern the static gravitational field sourced by the mass, while the solenoidal equations describe the propagating radiation field. This separation allows us to analyze the radiation field independently of the static gravitational component.

We can make this explicit by defining a new current:

$$\mathbf{J}'_S = -4\pi G \mathbf{J}_S \quad (23)$$

Since the solenoidal current is not constrained by the continuity equation (which governs only the irrotational subsystem), this redefinition carries no physical consequences. It transforms the inhomogeneous solenoidal equation into:

$$\nabla \times \mathbf{B}_S - \partial_t \mathbf{E}_S = \mathbf{J}'_S \quad (24)$$

while leaving all other equations unchanged. Dropping the prime, we adopt the following as the solenoidal (radiation) Maxwell equations:

$$\begin{aligned} \nabla \times \mathbf{B}_S - \partial_t \mathbf{E}_S &= \mathbf{J}_S \\ \nabla \times \mathbf{E}_S + \partial_t \mathbf{B}_S &= 0 \end{aligned} \quad (25)$$

We choose this normalization so that the effective Ohm's law (Section 6.2) takes its simplest form.

4.2 Spherical Vector Harmonics

Due to the radial symmetry of most problems in gravity, we will benefit from using vector spherical harmonics.

Following Barrera et al. [6] we assume that any vector field, $\mathbf{V}$, and any scalar, g , can be expanded as follows:

$$\begin{aligned} \mathbf{V}(\mathbf{x}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l [a_{lm}(r) \mathbf{Y}_{lm}(\theta, \phi) + b_{lm}(r) \hat{\mathbf{r}} \times \mathbf{X}_{lm}(\theta, \phi) \\ + c_{lm}(r) \mathbf{X}_{lm}(\theta, \phi)] \end{aligned} \quad (26)$$

$$g(\mathbf{x}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l g_{lm}(r) Y_{lm}(\theta, \phi) \quad (27)$$

where Y_{lm} are the usual spherical harmonics (see Jackson 3.53), $\mathbf{Y}_{lm}$ is simply $\hat{\mathbf{r}} Y_{lm}$ and $\mathbf{X}_{lm}$ is defined as,

$$\mathbf{X}_{lm}(\theta, \phi) = \frac{-i}{\sqrt{l(l+1)}} \mathbf{r} \times (\nabla Y_{lm}(\theta, \phi)) \quad (28)$$

The orthogonality relations and curl identities for these vector harmonics, which we use throughout, can be found in Barrera et al. [6] and Jackson [7] Chapter 9.

4.3 Irrotational Solutions

We now recover Newtonian gravity as a solution to the irrotational part of the Maxwell equations. For a static point mass m at the origin,

$$\rho(\mathbf{x}) = m\delta^3(\mathbf{x}), \quad \mathbf{J} = 0. \quad (29)$$

The irrotational equations are

$$\nabla \cdot \mathbf{E}_I = -4\pi G\rho, \quad \nabla \times \mathbf{E}_I = 0. \quad (30)$$

Expanding $\mathbf{E}_I$ in vector spherical harmonics and following the standard multipole expansion (see e.g. Jackson §3.9), the exterior solution is

$$\mathbf{E}_I(\mathbf{x}) = \sum_{l,m} \frac{-4\pi G q_{lm}}{(2l+1)r^{l+2}} \left[(l+1) \mathbf{Y}_{lm} + i\sqrt{l(l+1)} (\hat{\mathbf{r}} \times \mathbf{X}_{lm}) \right], \quad (31)$$

where the multipole moments are

$$q_{lm} \equiv \int Y_{lm}^*(\theta, \phi) r^l \rho(\mathbf{x}) d^3x. \quad (32)$$

For a point source, only $q_{00} = m/\sqrt{4\pi}$ is nonzero, and this reduces to

$$\mathbf{E}_I(\mathbf{x}) = -\frac{Gm}{r^2} \hat{\mathbf{r}}, \quad (33)$$

as expected. This confirms that Newtonian gravity is recovered as a special case of the irrotational Maxwell equations, a necessary consistency check for the framework. Note that while the point mass idealization suffices here, in **Section 5** we refine this to a shell distribution (equation 51) to address the gravitational self-energy problem.

4.4 Solenoidal Solutions

In the last subsection, we derived Newtonian gravity from the irrotational equations; in this subsection we derive the equations of radiation from the solenoidal solutions. Several results obtained here will be needed later.

Outside the mass, the solenoidal field equations in vacuum are:

$$\nabla \times \mathbf{E}_S + \partial_t \mathbf{B}_S = 0, \quad \nabla \times \mathbf{B}_S - \partial_t \mathbf{E}_S = 0 \quad (34)$$

We write the complete multipole expansion including both electric and magnetic type modes:

$$\begin{aligned}
\mathbf{B}_+ &= \int d\omega \sum_{l,m} \left[a_E(l,m) W_{lm}^E(\omega) f_l(kr) \mathbf{X}_{lm} \right. \\
&\quad \left. - \frac{i}{k} a_M(l,m) W_{lm}^M(\omega) \nabla \times (g_l(kr) \mathbf{X}_{lm}) \right] e^{-i\omega t} \\
\mathbf{E}_+ &= \int d\omega \sum_{l,m} \left[\frac{i}{k} a_E(l,m) W_{lm}^E(\omega) \nabla \times (f_l(kr) \mathbf{X}_{lm}) \right. \\
&\quad \left. + a_M(l,m) W_{lm}^M(\omega) g_l(kr) \mathbf{X}_{lm} \right] e^{-i\omega t}
\end{aligned} \tag{35}$$

with $\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_+^*$ and $\mathbf{B} = \mathbf{B}_+ + \mathbf{B}_+^*$. The functions f_l and g_l are linear combinations of spherical Hankel functions $h_l^{(1)}(kr)$ and $h_l^{(2)}(kr)$, and $W_{lm}^E(\omega)$ and $W_{lm}^M(\omega)$ are independent complex Gaussian stochastic variables satisfying:

$$\langle W_{lm}^E(\omega) \rangle = 0, \quad \langle W_{lm}^E(\omega) W_{l'm'}^{E*}(\omega') \rangle = \delta_{ll'} \delta_{mm'} \delta(\omega - \omega') \tag{36}$$

and identically for W_{lm}^M , with $\langle W_{lm}^E W_{l'm'}^{M*} \rangle = 0$. This stochastic noise should be viewed as the natural extension of the noise introduced in **Section 3** for spherical coordinates.

For a massive particle absorbing incoming radiation, we set $f_l(kr) = g_l(kr) = h_l^{(2)}(kr)$, since $h_l^{(2)}(kr) e^{-i\omega t} \sim e^{-i(kr+\omega t)}/(kr)$ at large r , representing ingoing waves.

Since there is no physical reason to prefer electric-type over magnetic-type modes in the vacuum, we impose the symmetry condition:

$$|a_E|^2 = |a_M|^2 \equiv |a|^2. \tag{37}$$

4.4.1 Computing the Field Momentum

The total field momentum is:

$$\mathbf{P}_{\text{field}} = \frac{1}{4\pi} \int \mathbf{E} \times \mathbf{B} d^3x \tag{38}$$

Writing $\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_+^*$ and $\mathbf{B} = \mathbf{B}_+ + \mathbf{B}_+^*$, the terms that survive the stochastic average are:

$$\langle \mathbf{E} \times \mathbf{B} \rangle = 2 \text{Re} [\langle \mathbf{E}_+ \times \mathbf{B}_+^* \rangle] \tag{39}$$

The independence of W^E and W^M eliminates cross terms between electric and magnetic modes. Working through the angular integrals using the orthogonality of $\mathbf{X}_{lm}$, $\hat{\mathbf{r}} \times \mathbf{X}_{lm}$, and $\mathbf{Y}_{lm}$, and combining the electric and magnetic contributions (which enter symmetrically under $|a_E|^2 = |a_M|^2$), the radial component of the Poynting-type integrand involves the Wronskian of the spherical Hankel functions.

Using the standard result $\mathcal{W}[j_l, n_l](x) = 1/x^2$ with the chain rule $d/dr = k d/dx$, one obtains:

$$h_l^{(2)} \frac{d}{dr}(r h_l^{(1)}) - h_l^{(1)} \frac{d}{dr}(r h_l^{(2)}) = \frac{2i}{kr} \quad (40)$$

Assembling all factors:

$$\hat{\mathbf{r}} \cdot \langle \mathbf{E} \times \mathbf{B} \rangle = - \sum_{l,m} \int d\omega \frac{4|a|^2}{\omega^2 r^2} |\mathbf{X}_{lm}|^2 \quad (41)$$

and the total field momentum is:

$$\langle \mathbf{P}_{\text{field}} \rangle = -\hat{\mathbf{r}} \int d^3x \int d\omega \sum_{l,m} \frac{|a|^2}{\pi \omega^2 r^2} |\mathbf{X}_{lm}|^2 \quad (42)$$

The negative sign confirms that the momentum is directed inward, as expected for absorbed incoming radiation.

Applying the hypothesis (1) and converting to frequency space using $d^3k = 4\pi\omega^2 d\omega$, we identify:

$$\sum_{l,m} |a|^2 |\mathbf{X}_{lm}|^2 = \frac{Gm\hbar\omega^4}{2\pi} \quad (43)$$

so that the total field momentum takes the form:

$$\langle \mathbf{P}_{\text{field}} \rangle = -\hat{\mathbf{r}} \frac{1}{8\pi^3} \int d^3x \int d^3k \frac{Gm\hbar}{r^2} \quad (44)$$

4.4.2 Field Energy in the Radiation Zone

The total field energy density is:

$$u = \frac{1}{8\pi} (\mathbf{E}^2 + \mathbf{B}^2) \quad (45)$$

In the radiation zone ($kr \gg 1$), the Hankel functions simplify to $h_l^{(2)}(kr) \approx \frac{(+i)^{l+1}}{kr} e^{-ikr}$, and the curl of $h_l^{(2)} \mathbf{X}_{lm}$ reduces to $-\hat{\mathbf{r}} \times \mathbf{X}_{lm}$ times the same scalar factor. The stochastic averages $\langle \mathbf{E}^2 \rangle$ and $\langle \mathbf{B}^2 \rangle$ each receive equal contributions from electric and magnetic modes, and cross terms vanish both by orthogonality ($\mathbf{X}_{lm} \cdot (\hat{\mathbf{r}} \times \mathbf{X}_{lm}) = 0$) and by the independence of W^E and W^M . The result is:

$$\langle \mathbf{E}^2 + \mathbf{B}^2 \rangle = \frac{8|a|^2}{\omega^2 r^2} \int d\omega \sum_{l,m} |\mathbf{X}_{lm}|^2 \quad (46)$$

giving the total field energy:

$$\langle U \rangle = \int d^3x \int d\omega \sum_{l,m} \frac{|a|^2}{\pi \omega^2 r^2} |\mathbf{X}_{lm}|^2 \quad (47)$$

Or, applying the hypothesis:

$$\langle U \rangle = \frac{1}{8\pi^3} \int d^3x \int d^3k \frac{Gm\hbar}{r^2} \quad (48)$$

The expansion of this calculation outside the radiation zone involves more intricate mathematics. We defer this more general treatment to future work.

5 A Toy Model

We now step back from the field-theoretic machinery to construct a simple model of the particle itself. Sections 3 and 4 laid out the two assumptions of the theory, both grounded in existing literature. From here, we enter new territory, making several idealizing assumptions in order to gain traction on the problem. Where a choice must be made, we take the simplest option. Less idealized models are left for future work.

We proceed in two steps. First, working in the space domain, we follow Rohrlich to establish a lower bound on the particle radius. Then, moving to the frequency domain, we introduce a single-frequency absorption model and show that the energy absorbed from the ZPR field is less than what is available, confirming the energetic consistency of the hypothesis.

5.1 Lower Bound: Minimum Radius via Gravitational Self-Energy

We follow Rohrlich[8] closely here, as his framing captures the essential point precisely. The passage has been lightly adapted from the original to reflect that we are discussing massive particles rather than charged ones.

We wish to construct a model of a massive particle. “The most obvious model of a massive particle is a sphere carrying a spherically symmetrical mass distribution. While such a model is meant (and was indeed proposed) as a picture of a massive elementary particle (a neutron, for example), it is obvious that it is basically a macroscopic massive body, only much smaller. There is nothing ‘elementary’ about it. To see this clearly, let us consider a macroscopic massive sphere in more detail.

“Consider a sphere of radius R and spherically symmetric mass density $\rho(\mathbf{x})$. Its total mass is

$$m = \int_V \rho(\mathbf{x}) d^3x \quad (49)$$

“The integral extending over the volume of the sphere. Since $m \neq 0$ by assumption, there must exist attractive gravitational forces between ‘parts’ of the total mass. Symmetry would therefore require that this massive sphere contract and continue to do so to arbitrarily small radius R .” Thus with no other forces present, the particle will necessarily collapse.

For the purposes of this paper, the mass distribution is taken to be a uniform spherical shell at radius R :

$$\rho(\mathbf{x}) = \frac{m}{4\pi R^2} \delta(r - R) \quad (50)$$

For brevity, we write this as:

$$\rho(\mathbf{x}) = m \delta_{\text{shell}}(\mathbf{x}) \quad (51)$$

In **Section 6**, we show that the particle need not collapse. For now, we take it as given that it does not, and ask what gravitational self-energy implies for the radius.

Since no forces other than gravity are present, the total energy satisfies:

$$U = -\frac{1}{2} \int \frac{Gm\rho(\mathbf{x})}{r} d^3x \quad (52)$$

Since the self-energy is negative, we use the relativistic statement $U^2 = m^2$ (equivalently, $p_\mu p^\mu = m^2$), which allows $U = -m$, giving:

$$m = \frac{1}{2} \int \frac{Gm\rho(\mathbf{x})}{r} d^3x \quad (53)$$

Choosing the simplest mass distribution, a uniform spherical shell at radius R , the self-energy integral yields $R = Gm/2$. Other choices (a uniform sphere, for example) give R of the same order. We therefore take $R = Gm$ as the characteristic minimum radius, absorbing numerical prefactors of order unity. We apply the same reasoning in **Section 6** when similar factors arise:

$$\boxed{R = Gm} \quad (54)$$

This is analogous in spirit to the classical electron radius, where self-energy sets a length scale.

5.1.1 Compactness of Physical Objects

It is illuminating to compare the mathematical radius with the radii of real physical bodies. Slipping back to SI units, we tabulate the compactness parameter $\Phi = \frac{Gm}{Rc^2}$ for several objects:

Mathematical limit: $\Phi = 1$

Black hole: $\Phi = 1$

Sun: $\Phi = 2 \times 10^{-6}$

Earth: $\Phi = 7 \times 10^{-10}$

Proton: $\Phi = 1 \times 10^{-38}$

Notably, the mathematical limit $\Phi = 1$ coincides with the black hole threshold, suggesting that $R = Gm$ is not merely a formal bound but a physically meaningful one.

5.2 Frequency Domain: The Single-Mode Approximation

Boyer showed rigorously that a dipole oscillator in the ZPR field absorbs from a remarkably narrow frequency band [9]; our single-frequency model can be understood as the idealized limit of this result. Any real model of a macroscopic massive object that absorbs radiation will absorb over a non-singular domain. However, to gain traction, we adopt a toy model that absorbs at only one frequency. We choose:

$$k_o = \omega_o = \frac{m}{\hbar} \quad (55)$$

In spherical coordinates, this corresponds to the frequency distribution:

$$\frac{\delta\left(\omega - \frac{m}{\hbar}\right)}{4\pi\omega^2} \quad (56)$$

The energy absorbed from the field, using (16), is then:

$$U = \int d\omega \hbar\omega \delta\left(\omega - \frac{m}{\hbar}\right) = m \quad (57)$$

To show that the particle draws less energy than is available from the ZPR, we introduce a multiplicative factor α . The self-energy condition gives $U = Gm^2/R$, and the absorbed energy is:

$$U = \alpha \int d\omega \hbar\omega \delta\left(\omega - \frac{m}{\hbar}\right) \quad (58)$$

Combining these:

$$\alpha = \frac{Gm}{R} \quad (59)$$

For any physical object with $R > Gm$, we have $\alpha < 1$: the particle absorbs less energy than is available. At the limiting case $R = Gm$, $\alpha = 1$ and the particle absorbs all available energy at that mode. The toy model is energetically consistent.

5.3 On-Shell Conditions

Before moving on, we collect the defining quantities of the model and show they are mutually consistent. We have the radius (54), the absorbed frequency, and the hypothesis (1) evaluated at $r = R$:

$$R = Gm, \quad k_o = \frac{m}{\hbar}, \quad p = \frac{Gm\hbar}{R^2} \quad (60)$$

Combining these with $p = \hbar k_o$ yields:

$$\frac{Gm\hbar}{R^2} = \hbar k_o = m \quad (61)$$

Multiplying through by $R/\hbar$:

$$k_o R = \frac{Gm}{R} = 1 \quad (62)$$

and therefore:

$$\boxed{k_o^2 R^2 = 1} \quad (63)$$

This result places the model exactly at the boundary between the near and far field regimes — neither $k_o R \ll 1$ nor $k_o R \gg 1$. We refer to this as being *on-shell*. Throughout what follows, $\delta_{\text{shell}}^{(3)}(\mathbf{x})$ denotes that we never probe scales smaller than R .

Similarly, $\delta_{\text{shell}}^{(3)}(\mathbf{k})$ denotes restriction to the single absorbed mode at $k_o = m/\hbar$, corresponding to the frequency distribution $\frac{\delta(\omega - m/\hbar)}{4\pi\omega^2}$ introduced above.

These relations are not imposed independently; each follows from the physical requirements of the model, and their mutual consistency is a nontrivial check.

6 Field Energy and Momentum Balance: A Mechanism for Stability

Now that we have established our model of a massive particle in the previous section, namely:

$$R = Gm, \quad m = \hbar k_o, \quad k_o R = 1 \quad (64)$$

we are ready to examine whether the proposed inflow of zero-point radiation can stabilize a massive particle against gravitational collapse. The instability in question was introduced in Section 5: a self-gravitating mass has divergent self-energy and will collapse unless some mechanism intervenes. We propose that the inflow of ZPR radiation provides exactly this mechanism.

Following Rohrlich, if the energy absorbed from the ZPR can offset the divergence in the gravitational self-energy, the particle is stable. We first analyze the energy density and total energy of the system, then examine whether this balance leads to equilibrium. Finally, we turn to momentum conservation, where we derive an effective “Ohm’s law” relating solenoidal fields and currents.

6.1 Stability from Field Energy

The irrotational contribution is:

$$U_I = \frac{-1}{8\pi G} \int |\mathbf{E}_I|^2 d^3x \sim \int \frac{Gm^2}{r^4} d^3x \quad (65)$$

This integral diverges, confirming that a massive particle would collapse under its own gravitational self-energy without a stabilizing mechanism. The radiation field provides precisely this stabilization. Including both contributions, the total energy is:

$$U_{\text{tot}} = \int_V \left(-\frac{1}{8\pi G} |\mathbf{E}_I|^2 + |\mathbf{E}_S|^2 + |\mathbf{B}_S|^2 \right) d^3x \quad (66)$$

From equation (48), the radiation contribution is:

$$U_{\text{rad}} = \frac{1}{8\pi^3} \int \int \frac{Gm\hbar}{r^2} d^3x d^3k$$

Integrating over k and suppressing numerical prefactors, which are absorbed into the definition of the model (as in **Section 5**), the total energy becomes:

$$U_{\text{tot}} = \int \left[\frac{Gm\hbar}{r^2} k^3 - \frac{Gm^2}{r^4} \right] d^3x \quad (67)$$

Evaluating on-shell, with $k_o = m/\hbar$, $k_o^2 R^2 = 1$ and $r = R = Gm$, both terms are equal and the integrand vanishes:

$$U_{\text{tot}} = 0 \quad (68)$$

The total energy vanishes on-shell, demonstrating that the radiation inflow exactly counterbalances the gravitational self-energy. The massive particle is stable.

6.2 Momentum Conservation and Ohm's Law

We now turn to momentum conservation. The total momentum exchange in the system must vanish:

$$\int_V (\rho \mathbf{E}_I + \mathbf{J}_S \times \mathbf{B}_S) d^3x = 0 \quad (69)$$

We assume the solenoidal current obeys Ohm's law:

$$\mathbf{J}_S = \sigma \mathbf{E}_S \quad (70)$$

Since the gravitational source is localized at the shell and the toy model absorbs at a single frequency, the conductivity must be localized in both position space and k -space:

$$\sigma \propto \delta_{\text{shell}}^{(3)}(\mathbf{x}) \delta_{\text{shell}}^{(3)}(\mathbf{k}) \quad (71)$$

With this, $\langle \mathbf{J}_S \times \mathbf{B}_S \rangle = \sigma \langle \mathbf{E}_S \times \mathbf{B}_S \rangle$. Using the result from equation (44), and absorbing numerical factors as we have done previously (see (54)), the radial component of the stochastic average of the Poynting-type integrand is:

$$\hat{\mathbf{r}} \cdot \langle \mathbf{E}_S \times \mathbf{B}_S \rangle = - \int d^3k \frac{Gm\hbar}{r^2} \quad (72)$$

The gravitational force density is also directed inward:

$$\rho \mathbf{E}_I = -m \delta_{\text{shell}}^{(3)}(\mathbf{x}) \frac{Gm}{r^2} \hat{\mathbf{r}} \quad (73)$$

For these to balance, σ must be negative: the current opposes the field, representing absorption. Evaluating on-shell, the factors of $1/r^2$ cancel and the integrals collapse, leaving:

$$\sigma = -\frac{m}{\hbar} \delta_{\text{shell}}^{(3)}(\mathbf{x}) \delta_{\text{shell}}^{(3)}(\mathbf{k}) \quad (74)$$

This defines an effective Ohm's law in which the mass acts as a point-like absorber of incoming radiation at a single frequency. The solenoidal Maxwell equation (using the normalization adopted in Section 4.1) becomes:

$$\nabla \times \mathbf{B}_S - \partial_t \mathbf{E}_S = -\frac{m}{\hbar} \mathbf{E}_S \delta_{\text{shell}}^{(3)}(\mathbf{x}) \delta_{\text{shell}}^{(3)}(\mathbf{k}) \quad (75)$$

Together, the energy and momentum analyses of this section demonstrate that the hypothesis is internally consistent: a massive particle that absorbs zero-point radiation is stabilized against gravitational collapse by drawing energy from the vacuum field itself.

7 Classical Tests of Gravity

This section demonstrates that the present theory reproduces the three classic experimental tests of general relativity, first proposed by Einstein [10]: the precession of Mercury's perihelion, the deflection of light near a massive object, and gravitational redshift.

7.1 Precession of the Perihelion of Mercury

In electrodynamics, a static Coulomb field transforms under Lorentz boosts into a configuration with both electric and magnetic components. Specifically, the fields of a charge moving at velocity $\mathbf{v}$ are:

$$\mathbf{E}' = \gamma \mathbf{E} - \frac{\gamma^2}{\gamma + 1} \mathbf{v}(\mathbf{v} \cdot \mathbf{E}), \quad \mathbf{B}' = -\gamma \mathbf{v} \times \mathbf{E} \quad (76)$$

where $\gamma = 1/\sqrt{1 - v^2}$.

By the assumption laid out in Section 4, we assume that the same transformation applies to the gravitational field

$$\mathbf{E}_g = -GM\mathbf{r}/r^3. \quad (77)$$

This is simply the irrotational field $\mathbf{E}_I$ of Section 4; we use the subscript g here to distinguish the gravitational field from the electromagnetic analogy.

In the low-velocity limit $v \ll 1$, we find:

$$\mathbf{B}_g = -\mathbf{v} \times \mathbf{r} \cdot \frac{GM}{r^3} \quad (78)$$

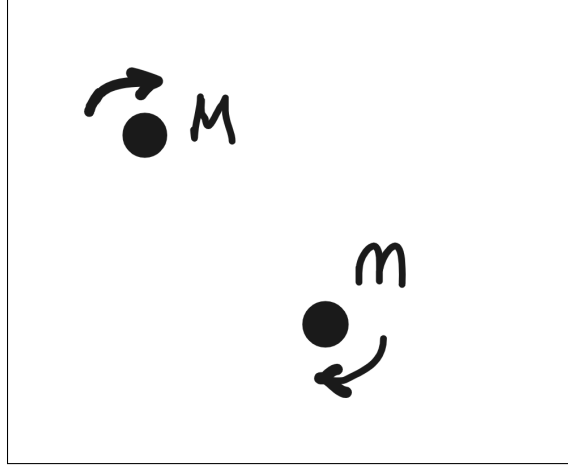


Figure 1: Two masses in mutual orbit

This introduces a "gravitomagnetic" interaction analogous to magnetism in electrodynamics.

We now consider Kepler's problem. Here M denotes the central mass and m the orbiting body, in contrast to the single-mass notation used elsewhere in the paper. The Newtonian effective potential is:

$$U_{\text{Newt}} = \frac{1}{2}m \left(\frac{dr}{dt} \right)^2 - \frac{GMm}{r} + \frac{l^2}{2mr^2} \quad (79)$$

where $l = m|\mathbf{r} \times \mathbf{v}|$ is the orbital angular momentum (see Figure 1).

A moving charge induces a magnetic moment $\mathbf{m} = \frac{1}{2}e(\mathbf{r} \times \mathbf{v})$. Replacing $e \rightarrow m$, we define a gravitational magnetic moment:

$$\mathbf{m}_g = \frac{1}{2}m(\mathbf{r} \times \mathbf{v}) \quad (80)$$

The interaction energy between this gravitomagnetic moment and a gravitomagnetic field $\mathbf{B}_g$ is:

$$U = -\mathbf{m}_g \cdot \mathbf{B}_g = -\frac{GMm}{2} \frac{|\mathbf{r} \times \mathbf{v}|^2}{r^3} \quad (81)$$

This adds a correction to the potential energy. Including the reciprocal effect (from the motion of the other mass), we double this term and find the corrected potential:

$$U_{\text{Einstein}} = \frac{1}{2}m \left(\frac{dr}{dt} \right)^2 - \frac{GMm}{r} + \frac{l^2}{2mr^2} - \frac{GMl^2}{mr^3} \quad (82)$$

This is the same correction obtained in general relativity to first order. As Carroll[11] notes, general relativity produces this result exactly; in our theory,

it emerges as a leading-order approximation. The higher-order terms in our expansion are not present in general relativity, and in principle offer an observational test that could distinguish between the two frameworks. We leave the computation of these higher-order terms for future work.

7.2 Deflection of Light

We now consider the deflection of light near a massive object. Recall our hypothesis that mass creates an inward-pointing spectral momentum density:

$$\mathbf{P}(r) = -\hat{\mathbf{r}} \frac{Gm\hbar}{r^2} \quad (1)$$

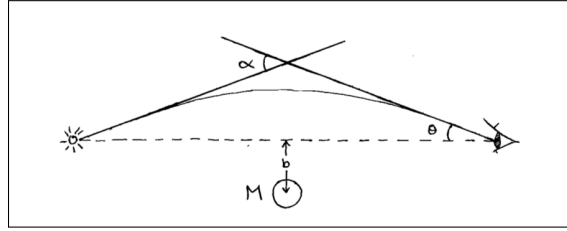


Figure 2: Deflection of light

A photon traveling through this field receives a cumulative momentum shift:

$$\Delta \mathbf{p} = \int_{\text{path}} \mathbf{P}(r) d\phi \quad (2)$$

where $d\phi = k dz$ is the infinitesimal phase along the path. To first order, we assume the photon follows a straight-line path at minimum distance b from the massive object (see Figure 2):

$$\Delta \mathbf{p} = - \int_{-\infty}^{\infty} \frac{Gm\hbar}{r^3} \mathbf{r} k dz \quad (83)$$

The parallel momentum shifts cancel symmetrically. The transverse component yields:

$$\Delta p_y = \hbar k \cdot \frac{2Gm}{b} \quad (84)$$

The angle of deflection is then:

$$\theta \approx \frac{\Delta p_y}{p_z} = \frac{2Gm}{b} \Rightarrow \alpha = \frac{4Gm}{b} \quad (85)$$

This matches, to first order, the deflection angle predicted by general relativity [12].

7.3 Gravitational Redshift

Finally, we consider the redshift of light climbing out of a gravitational well. A photon escaping radially outward from $r = R$ to infinity accumulates a momentum shift along its path. Since $\mathbf{P}(r)$ points inward and the photon travels outward, the momentum transfer opposes the photon's motion:

$$\Delta p = \int_R^\infty \left(-\frac{Gm\hbar}{r^2} \right) k dr = -\frac{Gm\hbar k}{R} \quad (86)$$

where $d\phi = k dr$ is the infinitesimal phase as before. The resulting frequency shift is:

$$\hbar k' = \hbar k + \Delta p = \hbar k \left(1 - \frac{Gm}{R} \right) \quad (87)$$

That is, a photon escaping a gravitational well loses energy, arriving at infinity with a lower frequency than it had at emission. This matches, to first order, the gravitational redshift predicted by general relativity [12].

8 Duality and the Dirac Equation

Having established that the theory reproduces the classical tests of general relativity, we now ask whether the field equations themselves contain any deeper structure. Throughout this paper, we have stayed grounded in classical physics. We built a theory of gravity using field equations, conservation laws, and an underlying sea of vacuum radiation. No quantum assumptions, no curved space-time. In this section, something surprising appears. When the field equations are rewritten in a new form, a familiar structure emerges: the Dirac equation, the cornerstone of quantum mechanics.

8.1 Duality Transformations

As discussed in Jackson [7] (Section 6.11), the Maxwell equations can be extended to include both electric and magnetic charges:

$$\begin{aligned} \nabla \times \mathbf{B} - \partial_t \mathbf{E} &= \mathbf{J}_e, & \nabla \times \mathbf{E} + \partial_t \mathbf{B} &= -\mathbf{J}_m \\ \nabla \cdot \mathbf{E} &= \rho_e, & \nabla \cdot \mathbf{B} &= \rho_m \end{aligned} \quad (88)$$

The magnetic densities are assumed to obey the same continuity equation as their electric counterparts.

Now consider the following duality transformation:

$$\begin{aligned} \mathbf{E} &= \mathbf{E}' \cos \xi + \mathbf{B}' \sin \xi, & \mathbf{B} &= \mathbf{B}' \cos \xi - \mathbf{E}' \sin \xi \\ \rho_e &= \rho'_e \cos \xi + \rho'_m \sin \xi, & \rho_m &= \rho'_m \cos \xi - \rho'_e \sin \xi \\ \mathbf{J}_e &= \mathbf{J}'_e \cos \xi + \mathbf{J}'_m \sin \xi, & \mathbf{J}_m &= \mathbf{J}'_m \cos \xi - \mathbf{J}'_e \sin \xi \end{aligned} \quad (89)$$

This transformation leaves the generalized Maxwell equations invariant. Note that these equations use the normalization for the solenoidal current adopted in Section 4.1, in which the factor of 4π has been absorbed into the definition of $\mathbf{J}$

8.2 Spinor Representation of Field Operators

We now examine a way to recast Maxwell's equations using spinor-like objects. Define the vector curl as a matrix operator:

$$\nabla \times \mathbf{V} = g_i \partial_i \mathbf{V} \quad (90)$$

with:

$$g_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad g_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad g_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (91)$$

We then define the gamma matrices as:

$$\gamma^0 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & g_i \\ -g_i & 0 \end{pmatrix} \quad (92)$$

These allow us to write the solenoidal Maxwell equations in a compact spinor form.

8.3 Emergence of the Dirac Equation

Recall from Section 6 that the on-shell momentum conservation equation (75) gives:

$$\nabla \times \mathbf{B}_S - \partial_t \mathbf{E}_S = -\frac{m}{\hbar} \mathbf{E}_S \delta_{\text{shell}}^{(3)}(\mathbf{x}) \delta_{\text{shell}}^{(3)}(\mathbf{k}) \quad (75)$$

Restricting to the on-shell condition (see Section 5), we drop the delta functions:

$$\nabla \times \mathbf{B}_S - \partial_t \mathbf{E}_S = -\frac{m}{\hbar} \mathbf{E}_S \quad (93)$$

The second solenoidal equation remains sourceless:

$$\nabla \times \mathbf{E}_S + \partial_t \mathbf{B}_S = 0 \quad (94)$$

At this stage, only the first equation carries a mass term. However, applying a duality transformation at $\xi = \pi/4$ mixes $\mathbf{E}_S$ and $\mathbf{B}_S$, and simultaneously generates a magnetic current $\mathbf{J}_m = -\frac{m}{\hbar} \mathbf{B}_S$ from the electric current $\mathbf{J}_e = -\frac{m}{\hbar} \mathbf{E}_S$. The two solenoidal equations become:

$$\begin{aligned} -\partial_t \mathbf{E}_S + \nabla \times \mathbf{B}_S + \frac{m}{\hbar} \mathbf{E}_S &= 0 \\ -\partial_t \mathbf{B}_S - \nabla \times \mathbf{E}_S + \frac{m}{\hbar} \mathbf{B}_S &= 0 \end{aligned} \quad (95)$$

Using the spinor curl definition:

$$\begin{aligned} -\partial_t \mathbf{E}_S + g_i \partial_i \mathbf{B}_S + \frac{m}{\hbar} \mathbf{E}_S &= 0 \\ -\partial_t \mathbf{B}_S - g_i \partial_i \mathbf{E}_S + \frac{m}{\hbar} \mathbf{B}_S &= 0 \end{aligned} \tag{96}$$

Defining a six-component field:

$$\psi = \begin{pmatrix} \mathbf{E}_S \\ \mathbf{B}_S \end{pmatrix} \tag{97}$$

these two equations combine into:

$$\left(\gamma^\mu \partial_\mu + \frac{m}{\hbar} \right) \psi = 0 \tag{98}$$

This has the exact form of the Dirac equation, as given in Sakurai's Advanced Quantum Mechanics[13], Eq. 3.31.

9 Conclusion

Can gravity emerge from classical physics, without invoking curved spacetime or quantum gravity? That was the question we asked. Faced with two powerful but incompatible theories, we chose a different path: to treat mass as a source of momentum flow, drawing from a pervasive background field. The result stands in contrast to general relativity, where mass is treated as a cold object sitting in empty space and curving spacetime around it. Here, there is heat in the universe, and that has made all the difference.

To test the idea, we built the theory step by step. We formulated equations for how mass interacts with the vacuum field. We constructed a toy model, choosing the simplest possible idealizations. We checked for conservation of energy and momentum. Then we asked: does this theory match observation? And the answer, to leading order, is yes. It reproduces the familiar predictions of general relativity: the bending of light, gravitational redshift, and the precession of Mercury. We also identify where the theory diverges from general relativity, suggesting a possible observational test.

We saw that when we re-expressed the field equations, we were able to derive the Dirac equation. That a single classical field framework can describe gravity, electromagnetism, and quantum behavior is remarkable. It suggests these are not separate layers of physics, but different faces of the same underlying structure. One set of equations that does it all.

9.1 Where This Could Go

Several directions remain open for future work:

Quantization. The Dirac equation presented here is not quantized, and the microscopic interaction mechanism is left unspecified. We suspect that a

two-photon singlet state may play the role of the interaction particle, but this remains speculative.

Derivation of the hypothesis. The intuition underlying the hypothesis has respectable precedent. Feynman notes that light “falls” in a gravitational field [1], which is precisely the picture developed here: mass acts as a sink for radiation. What is novel is the specific form of the momentum density, $\mathbf{P}(r) = -\hat{\mathbf{r}} Gm\hbar/r^2$. We suspect this expression can be derived directly from the two assumptions of the paper, rather than postulated. That derivation is left for future work.

Energy expansion outside the radiation zone. The energy calculation in Section 6 was carried out only in the radiation zone. Extending this to the full domain involves richer mathematics and, we suspect, richer physics.

Higher-order perihelion precession. The perihelion precession calculation was performed to leading order, matching general relativity. Higher-order terms in our expansion are not present in general relativity and could provide an observational test distinguishing the two frameworks.

Less idealized models. The toy model of a massive particle adopted here is as simple as possible. Less idealized models would be worth studying in future work.

Experimental test. We identified where the theory diverges from general relativity. Designing a concrete observational test is an open challenge.

Gravitational waves. The solenoidal sector of the theory naturally supports radiative solutions. Whether these match the properties of gravitational waves observed by LIGO, including propagation speed, polarization, and energy loss in binary systems, is an important open question.

9.2 Final Thought

Einstein advised making a theory as simple as possible, but no simpler. Bringing electricity, magnetism, gravity, and the Dirac equation under one set of equations is a level of simplicity that deserves serious attention.

The history of physics is full of bold guesses. Some are wrong, some are premature, and some are foundational. This is just a guess. But it is a specific, testable, and mathematically structured one. If nothing else, it shifts the question. Gravity need not emerge from geometry. It might instead arise from the quiet hiss of radiation no one thought to listen to.

This is offered not as a conclusion, but as a beginning. Collaborators welcome.

Acknowledgments

This paper would not exist without the support of a few key people. First and foremost, I thank my wife, who has stood beside me for over two decades as I pursued this idea. Her patience, steadiness, and quiet encouragement through all the ups and downs have been unwavering.

I am also deeply grateful to my friend Michael Seidel, who, despite not having a background in physics, listened to me explain these ideas again and again over the years. His willingness to engage, ask questions, and simply be present helped me find clarity when I needed it most.

To both of them: thank you.

References

- [1] R.P. Feynman, F.B. Morinigo, and W.G. Wagner, *Feynman Lectures on Gravitation*, edited by B. Hatfield (Westview Press, Boulder, CO, 1995).
- [2] T.H. Boyer, “Stochastic Electrodynamics: The Closest Classical Approximation to Quantum Theory,” *Atoms* **7**(1), 29 (2019).
- [3] T.H. Boyer, “Derivation of the Blackbody Radiation Spectrum without Quantum Assumptions,” *Physical Review* **182**, 1374–1383 (1969).
- [4] O. Heaviside, “A Gravitational and Electromagnetic Analogy,” *The Electrician* **31**, 281–282 (1893); reprinted in *Electromagnetic Theory*, vol. 1, pp. 455–466.
- [5] B. Mashhoon, “Gravitoelectromagnetism: A Brief Review,” arXiv:gr-qc/0311030 (2003).
- [6] R.G. Barrera, G.A. Estevez, and J. Giraldo, “Vector Spherical Harmonics and Their Application to Magnetostatics,” *European Journal of Physics* **6**, 287–294 (1985).
- [7] J.D. Jackson, *Classical Electrodynamics*, 3rd ed. (Wiley, New York, 1999).
- [8] F. Rohrlich, *Classical Charged Particles*, 3rd ed. (World Scientific, Singapore, 2007).
- [9] T.H. Boyer, “Random Electrodynamics: The Theory of Classical Electrodynamics with Classical Electromagnetic Zero-Point Radiation,” *Physical Review D* **11**, 790–808 (1975).
- [10] A. Einstein, “Die Grundlage der allgemeinen Relativitätstheorie,” *Annalen der Physik* **49**, 769–822 (1916).
- [11] S.M. Carroll, *Spacetime and Geometry: An Introduction to General Relativity* (Addison-Wesley, San Francisco, 2004).
- [12] J.B. Hartle, *Gravity: An Introduction to Einstein’s General Relativity* (Addison-Wesley, San Francisco, 2003).
- [13] J.J. Sakurai, *Advanced Quantum Mechanics* (Addison-Wesley, Reading, MA, 1967).